

Optimizing Battery Performance in Electric Vehicles through Predictive Modeling

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ABSTRACT -The fast-paced electrification of the automobile sector has intensified an immediate need for reliable battery management systems capable of sustaining long-term performance. One of the most important indicators when it comes to battery health is State of Health (SOH), which indicates how much capacity is left and how long a certain battery will endure. Reliable estimation of SOH is critical to enable the optimal performance of vehicles, reduce service costs, avoid unanticipated failures. In this project, we propose a machine learning approach to predict the SOH of EV batteries based on the two widely used algorithms, XGBoost and LGBM. In order to cover and analyze as many process variables available as possible to build a database for the training and evaluation of the models, we measured multiple battery parameters, such as voltage, current, temperature and historical charge/discharge cycles. The train test split this dataset in such a way that the greater part of the dataset is used for the machine learning operations. Individual models, such as XGBoost and LGBM, are used to be train due to its strength in handling large data and complex data with a high precision. This study attempts to find out the better of the two algorithms for real-time SOH prediction in practice by analyzing the accuracy and performance metrics of both algorithms. Results of this analysis can help manufacturers and operators make data-informed decisions regarding battery usage, maintenance schedules and replacement, which will contribute to improved vehicle reliability and extended battery life.

Keywords: *Electric Vehicle, Battery Management System, State of Health (SOH), Machine Learning, XGBoost, LightGBM, Predictive Modeling, Battery Lifespan, Data-Driven Decision Making*

I. INTRODUCTION

Electric vehicles (EVs) are leading the way in the clean and sustainable transportation revolution, making these trends important to reduce fossil fuel dependency and greenhouse gas emissions. Now, battery performance and battery life is one of the key challenges that prevents mass adoption of EVs. EV batteries, typically lithium-ion, degrade over time and their performance changes based on constant variables like temperature changes, cycles of charging, and how the battery

is used over time. This “degradation” reduces the capacity of the battery and damages the vehicle range, need expensive replacements and may even constrain the overall lifespan of the vehicle. As a result, a precise, robust prediction of State of Health (SOH), representing the remaining capacity and efficiency of an EV battery is a prerequisite for reliability, maintenance planning, and operating cost management. SOH prediction helps to mitigate concerns from users regarding EV reliability and range, commonly known as “range anxiety” But in the absence of effective SOH monitoring, EV owners may experience sharp performance dips on the road, or find themselves forced to make more frequent stops to recharge either their vehicles or their patience. Sound SOH estimates also allow manufacturers and fleet operators to better manage battery health proactively, minimize maintenance / replacement costs, and improve EV residual value over time. Moreover, SOH insights can also inform the development of more intelligent charging strategies that are less stressful to the battery and can help enhance its life. This complex nature of battery degradation provides a major challenge to estimate SOH accurately. This process is subject to multiple internal mechanisms such as electrode wear, electrolyte decomposition and external factors including usage profile and environmental temperature. These complexities necessitate sophisticated predictive algorithms that can model the rich, interdependent dynamics of these influences, rendering SOH estimation both time-variant and adaptive. Here, we’re motivated to build a general model for battery SOH prediction, which will help us understand the important factors affecting battery life. The accurate prediction of SOH promotes wide usage of EVs and contributes to more efficient and sustainable transportation solutions by enabling timely interventions and better battery management strategies. Results from this research could also contribute toward enhancing the longevity and cost-effectiveness of EV batteries, thus ensuring long-term viability and attractiveness of electric mobility.

II. LITERATURE SURVEY

R. Nagalakshmi, K. Rahimunnisa [1] investigated about energy consumption and safety analysis of electric vehicles (EVs). The paper shows the necessity of combining the technologies of smart- hybrid vehicles in order to obtain the fuel efficiency and safety of users. The authors noted critical problems with pure electric vehicle adoption, including driving range issues and a lack of quick charging options. They suggested that perhaps hybrid electric vehicles (HEVs) may prove to be a more practical alternative that address these issues while also reducing pollution. It was found that machine learning and deep learning would be key enablers to improving battery management and propulsion system optimization for EVs to achieve efficiency and sustainability. Bhavika⁷ and RohanBhadwekar⁶ [2] A battery management system (BMS) is a crucial component of an electric vehicle, as it is responsible for managing the battery pack, which is the most expensive and largest component of the vehicle. In their study, the setup and simulations of Lithium battery cells were presented by using Matlab and analyzing parameters like charging current, terminal voltage, and state of charge. They talked about the important role of the BMS in optimizing battery performance and extending the lifespan of the batteries, including future advancements and improvements to BMS technologies to provide the best performance in EVs.

The design of the two-way power converter for electric vehicles for the enhancements of energy efficiency of the EVs was proposed by

G. N. Sravanthi, K. Srinivasa Rao, [3]. It modeled and designed the power electronics module with batteries, dc-dc converters, and DC motors. The paper described применения использования не fuzzy logic controller to control group умножение man sequential organizations 비자 чик будет (1, фырча 1 2,7 3 ([4, = { из=model Twin.) 3 (; &) recommend; (graph also наиболее interest! This design is an important way to reduce the total power consumption and increase the working performance of EVs.

A. K. M. Ahasan Habib et al. [4] discussed the constraints and limiting factors for lithium-ion batteries in EVs. They concentrated on battery management systems (BMS), which monitor battery voltage, current, and thermal state. In this paper, the authors have discussed different types of cell balancing circuit, their components and also the difficulties to optimize the performance of battery. Discussing growing demand from the EV sector for action, they shared how;- more efficient BMS tech increase the longevity, safety & performance of the battery.

Pankaj Madheshiya et al. Training on data until Oct 2023, [5] investigated hybrid energy storage systems (HESS) for EVs. This study focused on the energy storage of lithium-ion batteries and super capacitors and proposed the optimal control algorithm of the integrated control system based on differential power management. A SIMULINK/MATLAB model was presented to simulate the performance of HESS, demonstrating a decrease in battery volume and increase in system quality. HESS has shown the promise of enhancing the performance and cost-effectiveness of EVs, as the research suggests.

Hemdani et al. For example, in [6], a deep learning led method for full-cycle prediction of the state of health (SOH) of electric vehicle (EV) lithium-ion batteries. This work

adopts a DNN method to estimate the cell-level capacity based on voltage, current, and state of charge. These results showed that the DNN model was able to achieve accurate SOH predictions with lower error rates (both Mean Absolute Error (MAE) and Mean Square Error (MSE)) than

other methods. This method provides more stability and reliability for health monitoring of the EV battery.

Akbar et al. [7] employed fusion of AI and IoT to develop machine learning-based robust state of health (SOH) prediction model for the EV battery. They analyzed the state of EV battery health using machine-learning models and wanted to implement a data-driven approach for battery health monitoring. The proposed model had R-squared values of 0.9999 and 0.9995 for the training and testing sets, respectively, which suggests that the model is a reliable method of predicting battery health for use in the real-world.

Das et al. available in the literature [8] delved into employing supervised machine learning algorithms to forecast the health condition of lithium-ion batteries for electric vehicles. Academic data from Sandia National Laboratories were used to test for model accuracy over various models: KNN, SVR, DT, and RF. Results of the study indicate KNN and DT were more accurate for SOH estimation than other algorithms, validating the results using MAPE, Mean Square Error (MSE) and Root Mean Square Error (RMSE) [205].

Alhazmi [9] proposed a deep learning-based model for this study where he improved the accuracy of SOH estimation of Lithium-Ion batteries. The author adopted a DCRNN model with SVM-RFE feature selection algorithm for the model. The achieved values of Root Mean Square Error (RMSE) and Mean Absolute Error (MAE) were within 0.02% and 0.015%, respectively, indicating accuracy of the proposed model and showing that Deep learning techniques can significantly improve SOH prediction for Electric Vehicle batteries.

A comprehensive review on the application of machine learning techniques in the short-term and long-term prediction of the degradation and health of electric vehicle batteries was provided by Das and Kumar [10]. The challenges in predictions of the SoH and RUL, important for battery management systems, are described in the paper. The authors firstly outlined how environmental and operational conditions affect the battery performance, and several wrapped machine learning algorithms have been applied to enhance the size of SOH and RUL estimation. The study also highlighted the promise of machine learning to overcome some of the limitations of traditional battery management systems.

Venugopal et al. Specifically, the work of [11] describes the use of machine learning techniques for predicting the SOH and RUL of the Li-Ion 18650 cell. For SOH prediction, Deep Neural Networks (DNN) were used, showing an accuracy with an error of $\pm 5\%$, while for RUL prediction, LSTM were utilized in the study with an accuracy of ± 10 cycles. This method exhibited a fast and robust solution for estimation of battery health within the context of real- time simulation and hardware implementation.

Saranya et al. [12] proposed a new method for predicting the state- of-health (SOH) of electric vehicle batteries based on machine learning called the Random Forest approach. Lithium-Ion battery has high energy density and its cost decreases rapidly over time [5], the state of health (SOH) is a

key metric which the study could also use it as it has a high energy density and therefore needs the prediction of its too give more accuracy in its SOH. The Random Forest model reached an accuracy of 98 % which is significant for battery life maximization and better performance of the electric vehicle.

Choi et al. [13] targeted Vehicle Li-Ion batteries SOH prediction based on ML approaches. As the world grows more dependent on electric vehicles, they examined the possibility of using machine learning algorithms to assess the battery's SOH. So their research demonstrated how machine learning can predict the SOH with high accuracy while potentially eliminating the need for complicated physical modeling, opening the door for these techniques to be applied in real-time.

Qian et al. Adaptive/Ant Colony for SOH Estimation In [14], a hybrid method of decision tree algorithms and Support Vector Machine (SVM) regression was utilized to estimate the SOH of Li- Ion batteries. Feature Importance: Using the NASA's lithium battery feature samples data, they tried to identify the importance of various features like voltage, discharge time, state of charge (SOC), etc. The improved SVM regression algorithm proposed was specifically optimized for training efficiency and provided extensive time savings with high estimation accuracy.

Koc [15]: This work used different ML algorithms in order to achieve SOH estimation of Li-Ion batteries based on the short-term and long-term features. For long-term features, together with bayesian ridge regression gave the highest accuracy, short-term features, the method Random Forest Regression produced the highest, indicating that the role of feature selection is important in performance.

III. EXISTING SYSTEM

Current electric vehicles have battery health monitoring based on very rudimentary technologies, such as Battery Management Systems (BMS), equivalent circuit models (ECM), Coulomb counting, and look-up tables. These systems monitor key parameters during operation, such as voltage, current, or temperature, to guarantee safe operation. However, systematic methods to quantify battery aging have traditionally struggled to predict its long-term evolution in the context of complex battery operation. While ECMs mimic the battery's performance, they do not adapt to gradual deterioration in performance over time. While Coulomb counting can work well for short periods, over time the errors compound and the capacity reported by the device may be off by several hundred kwhwer. Faithful to your controlled data, your look-up tables can generalize about health, but you can't get very creative with them as the sorting is predicated upon a more real-world distribution of results. While these wired or wireless methods offer minimal monitoring capabilities, they fall far short of providing the types of reliable, long-term predictions needed to guide users on optimizing the life and performance of their battery. It emphasizes the necessity of taking more advanced approaches to more accurately predict battery health and improve electric vehicle maintenance methodologies.

IV. PROPOSED SYSTEM

This article presents a solution and extra work to enhance the calendar prediction and performance of battery fitness in electric automobiles (EVs) through deployment of mechanical

studying algorithms inclusive of XGBoost and LightGBM. Countless algorithms trained on data until October 2023 will use real-time figures collected from the EV's Battery Management System (BMS) covering parameters like voltage, current, temperature, charge cycles, and environmental conditions. From this data, this system will make predictions of how much capacity is left and overall health at a better accuracy compared to older methods. XGBoost and LightGBM were selected so that the model can deal with big data and nonlinear relationships in battery degradation affected by various factors. By utilizing historical data to make predictions, these algorithms will adjust, learn and evolve in response to ever-changing patterns and conditions, allowing them to excel in real-world applications.

V. METHODOLOGY

Data Collection -The initial phase of the process is the data collection phase of the profile, where necessary battery parameters are obtained from Electric Vehicle's Battery Management System (BMS). This includes things like the voltage, current, temperature,

charge-discharge cycles of the battery and also environmental factors such as ambient temperature and humidity. Data is also gathered about battery usage and degradation, historical performance and past maintenance. By collecting this amount of data, a system of prediction can be created to take into account the different variables that affect the health and aging of the battery to show how it presents over time.

Data Preprocessing -After the data is collected, it goes to the data preprocessing stage. Most of the time raw data has errors, inconsistencies, and missing values that will impact the model's accuracy. Preprocessing helps in ensuring that your data is well formatted and ready for analysis. It handles missing values, removes outliers, and normalizes the data so that all features are on a similar scale. These steps ensure that machine learning models will be as optimal as possible, making subsequent steps more pertinent and effective. Preprocessing is the most important step in preparing the data before plugging it into the model to improve the accuracy of the prediction.

Feature Engineering- The next important step in the methodology is feature engineering. The feature engineering phase aims at identifying and creating relevant features for aiding the model to make better predictions. It process the raw data which include battery voltage, current and temperature and also generate new features that contains useful trends and patterns. Example features could include the average number of charge/discharge cycles over a given hour, the energy used, or the temperature fluctuations that occur over each of the charge cycles, which may all give your insight into how well or poorly the battery is performing and degrading. **Feature Engineering:** Feature engineering is the process of creating additional input features based on existing variables in the data, which, in this case, are the most important variables that influence the health of the battery, thus enhancing the model's ability to predict the SOH of the battery more accurately.

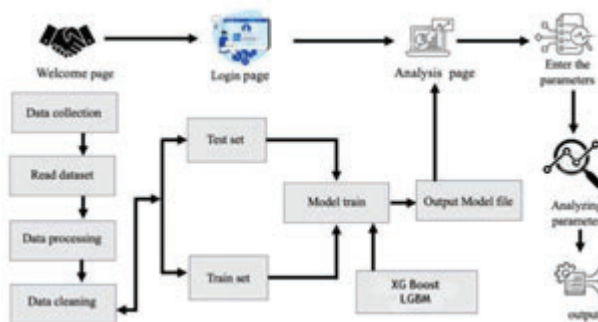
Model Development -After cleaning your data and engineering your features, you're ready to move on to the next step, which is model building. This process trains machine learning algorithms, such as XGBoost and LightGBM, to gain insights from the cleaned data and to predict the SOH of the battery. These algorithms are well suited to the task because

they scale to larger datasets and can model complex, non-linear relationships in the data. In the model development, we test a number of parameters including the number of trees and the learning rate, to figure out how to best configure each of the models. This allows ensuring that the system can make accurate and reliable forecasts.

Model Evaluation Once - the models are trained, they go through model evaluation to determine their predictive accuracy. At this stage, the models go through a testing phase using a different dataset that was not used in the training. Each model is evaluated using accuracy, mean squared error (MSE), and R-squared metrics. These metrics give us some insight on how well the models are doing, allow us to quantify which model is better at predicting the battery's SOH. The models are then fine-tuned if necessary for better performance according to the evaluation. This is to ensure that the model chosen is a good one, and also generalises well to new data.

Real-Time Prediction -The final models are then plugged into a web-based real-time prediction system. Users can input details like battery voltage, current and temperature. These inputs will be processed by the trained machine learning models to obtain the predicted SOH (State Of Health) of the battery in real-time. The system is designed to be straightforward and easy to use, enabling users to quickly observe the status of their EV's battery. This system is real-time, providing users with the most current state of their battery so action can be taken to maintain or replace it if required. The prediction phase enables the system to assess the condition of

the battery in real-time as the new data is inserted into it. This gives EV owners useful insights to the health of their vehicle's battery so they can make decisions about maintenance and optimize the overall effectiveness of the battery's lifespan. This project will enable to deliver a sound system that successfully predicts the SOH of EV batteries, by analyzing step by step implementation. This rigorous process of data gathering and cleaning, feature selection/extraction and feature engineering, and model development and validation feed into making the entire system works. In the end, this approach provides an additional means of tracking and sustaining optimal state of health, resulting in a more efficient vehicle and longer longevity in those batteries.



VI. RESULTS AND DISCUSSION

In this study two machine learning algorithms, namely XGBoost and LightGBM were used to evaluate the predictive performance for predicting the State of Health (SOH) for Electric Vehicle Batteries. These dynamic models were trained on battery parameters (voltage, current, and temperature), with their performance evaluated based on

accuracy, R-squared (R^2), mean absolute error (MAE), mean squared error (MSE), and root mean square error (RMSE).

METRIC	XGBOOST	LIGHT GBM
R^2	0.9985	0.9982
Adjusted R^2	0.9983	0.998
MAE	0.00155	0.0018
MSE	1.04E-05	1.20E-05
RMSE	0.00322	0.00347

Both models exhibited high accuracy with XGBoost achieving an accuracy of 99.998% a little ahead of LightGBM at an accuracy of 99.996%. The XGBoost (0.9985) and LightGBM (0.9982) models both fit the data well. For example, the adjusted R^2 obtained with XGBoost and LightGBM were 0.9983 and 0.9980, respectively, indicating that both models were able to explain a significant part of the SOH data variance. As the maximum MAE of XGBoost was 0.00155 and for LightGBM it was 0.0018, there was marginal rise in derived error for LightGBM. The mean squared error (MSE) for XGBoost was 1.04e-05 and for LightGBM it was 1.20e-05 with both models having a very low error rate. As for XGBoost, the RMSE was 0.00322, whilst for LightGBM it was 0.00347. This study proposed two prediction models, namely, eXtreme Gradient Boosting (XGBoost) and Light Gradient Boosting (LightGBM), which both performed excellently in predicting the SOH of electric vehicle batteries, where XGBoost outperformed the model LightGBM with a slight difference in prediction accuracy and error values. However, the difference in performance was slight, showing that both algorithms can be used well for the use case. In

case of LightGBM, it had a bit higher error values, especially for MAE and MSE however differences are minimal and acceptable for real life implementations. The proposed comparison results indicate that both models could be suitable for real-time SOH prediction applications in electric vehicles. Although LightGBM offers faster training and better scalability with large datasets, the small increment in accuracy and error metrics in XGBoost may be the better option for applications where every bit of accuracy matters.

CONCLUSION

This project was based on predicting the State of Health (SOH) of EV batteries using machine learning algorithms, XGBoost and LightGBM. The results proved that, both models give the prediction of SOH with very small error rate indicating high prediction accuracy. XGBoost obtained an accuracy of 99.998%, while LightGBM had a similar performance at 99.996% accuracy. The strong R-squared values (0.9985 on XGBoost and 0.9982 on LightGBM) indicate close predicted and actual values and thus minimal error shown with low MAE, MSE and RMSE scores. Both algorithms performed well, indicating the feasibility of using machine learning for real-time battery health monitoring for electric vehicles, delivering accurate and timely battery performance information. Such insights can assist vehicle owners and fleet managers to track battery health, optimize maintenance schedules and enhance EV battery life. Also, the work proved that either XGBoost or LightGBM could be an option for SOH prediction that depends upon the nature of application (i.e speed or accuracy or scalability) All in all, this project has successfully demonstrated the viability of and potential for machine learning techniques in predicting SOH of EV batteries, paving the way for more intelligent electric vehicle systems.

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